Image-based Encrypted Traffic Classification with Convolution Neural Networks

1st Yanjie He
Xi'an Jiaotong University
xian, China
2625520385@qq.com

2ndWei Li
Xi'an Jiaotong University
xian, China
liw@xanet.edu.cn

***Abstract*—Network traffic classification plays an important part in the network management and network monitoring. It can help administrators to understand the constitution of network traffic, facilitate administrators to manage the network and provide differentiated service quality and security monitoring. However, the widespread usage of the encryption techniques and dynamic ports policy make encrypted traffic classification become a great challenge for traditional traffic classification methods. In this paper, we propose an image-based method that can classify encrypted network traffic with a high accuracy. The basic idea of the method is to convert the first few non-zero payload sizes of session to gray images, and classify the converted gray images with convolutional neural network to achieve the goal of categorizing the encrypted network traffic. This method is very light-weight and it can automatically extract features, select features and classify encrypted network traffic to categories. We use the public dataset ISCX VPN-nonVPN to validate our proposed method. The experimental results show that our proposed approach achieves F1 score of 97.73% on the conventional encrypted traffic classification and F1 score of 99.55% on the virtual private network traffic(VPN) classification.**

***Index Terms*—Encrypted traffic classification, Image, CNN.**

I. Introduction

Network traffic classification is an important task for network management and network security. It can improve quality of network service and optimize the allocation of network resources. For example, in the network management, the network managers of enterprise and campus can know of the network traffic distribution over a period of time through network traffic classification, make some policies for adjusting network resources to guarantee an efficient management [1]. In the network security, network traffic classification can detect malicious use of network resources through knowing the distribution of network traffic. Network traffic classification can help network managers to identify the malicious system attacks [2].

According to the different granularity of network traffic, network traffic classification can be classified as network protocol classification, network traffic type classification and network application classification. The network protocol classification is to classify the network traffic into a specific protocol, such as SSL, DNS and SMTP [3] [4]–[6]. The network traffic type classification is to classify the network traffic into traffic types, such as Chat, Email and VoIP. For example, we categorize the traffic generated by two people communicating through Gmail application as Email type. We can use instant-messaging application Skype to communicate in many ways, such as text chat and voice chat. We categorize the traffic generated by two people communicating in the text way through Skype application as Chat type, and categorize the traffic generated by two people communicating in the voice way through Skype application as VoIP type [7], [8]. So far, there is only few researches on this classification. The network application classification is to group the traffic generated by a certain application as this application. For example, we use Skype for text chat, voice chat, video chat one after another, and we categorize the traffic and related traffic generated by these specific activities as Skype application [1], [9], [10]. In this paper, we classify network traffic to traffic type.

According to the characteristics of network traffic classification methods, these methods can be generally categorized as port-based methods, payload-based methods and statistical-feature-based methods. Port-based methods were effective and precise in the early days of the Internet, but with the widespread use of dynamic port policy and port disguise in many applications, port-based methods become more and more inaccurate to classify network traffic, whether unencrypted or encrypted. The wide usage of encrypted techniques in many applications makes the signature and information in the payload of packets unavailable. Thus the payload-based methods only classify the encrypted traffic with coarse granularity, such as SSL [11]. The statistical-feature-based methods commonly use machine learning algorithms to classify the network traffic. These methods can effectively classify encrypted network traffic, but they need to manually design features of network traffic (e.g. mean, standard deviation and inter-arrival time of flow) and select useful features via the trial-and-error pattern. Thus these methods not only need to take much time and energy to observe the sample data, but also overly depend on experience of experts [3]. Even if these methods use machine learning algorithm to select feature subset for encrypted traffic classification from a large set of features derived from the traffic data, this is at the expense of high computational cost and the feature set derived from the data is limited. Consequently, the selected feature subset for encrypted traffic classification is not necessarily optimal. Therefore, a machine learning algorithm that is capable of automatically extracting

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and selecting salient features from encrypted traffic is in urgent demand. Deep learning algorithm is a typical machine learning approach, it can automatically learn features from raw data and obtains remarkable achievements in many fields, such as computer vision and natural language process. Hence how to apply the deep learning algorithm to the encrypted traffic classification is a difficult problem to be solved. So far, the most end-to-end works focus on different deep learning algorithms for traffic classification, but ignore the optimization of encryped traffic itself [2], [12] [7], [13]. In this paper, we study the effect of length of subflow (the first few payload sizes of session) on classification performance.

To resolve these problems, in this paper, we propose an image-based method, this method belongs to end-to-end method. This method converts first few non-zero payload sizes of session to gray image and employs the deep learning architecture of one dimensional convolution neural network (1D-CNN) to automatically extract and select features and classify converted gray images to categories to achieve the purpose of classifying encrypted network traffic. The proposed method only use the first few non-zero payload sizes of each session to classify encrypted network traffic, thus this method can take low computational cost and is implemented at a high speed. The first few non-zero payload sizes of session, which are the handshake procedure of Transport Layer Security or Secure Sockets Layer (TLS/SSL) encryption protocol and a few content data, contain the intrinsic characteristics of sessions. The reason of this choice is given in the section V. And this choice is similar with [1], [2], the two use the characteristics of first N payloads of flows or sessions to classify the malicious traffic and application traffic. In addition, the proposed method does not need handcrafted features and rely on the experience of experts. The process of our proposed method is automatically collaborative optimization, the sub-process feature extraction and selection are not limited to finite manual feature set, thus the proposed method can extract and select salient features for encrypted traffic classification than statistical-feature-based methods. Finally, the process of proposed method can get global optimal solution. This process is also the process of scientific decision-making. The contributions of the paper are as follows:

- We achieve the length of subflow to improve the classification performance instead of using first 784 bytes or 900 bytes of session arbitrarily.
- Our method uses only first few non-zero payload sizes of each session for encrypted network classification, thus the method takes low computational cost and can be easily implemented.
- Our proposed method achieves significant improvement in classification performance both conventional traffic and VPN traffic.

Roadmap. Sec. II briefly presents the related works. Sec. III presents the knowledge on convolution neural network. Sec. IV describes the proposed methods. Sec. V mainly covers the experiment results and analysis. Sec. VI describes a few problems on the proposed method and future work. Sec. VII presents the conclusion.

II. Related Work

The study of network traffic classification has gone on in the past decades due to its essential value for achieving better quality of service (QoS), network security and network monitoring [12]. Researchers do lots of works on this topic and obtain many effective results. We categorize these methods into four categories base on their characteristics. The four categories are port-based approach, deep payload inspection (DPI) approach, statistical and machine learning approach and end-to-end method. In this section, we briefly introduce the characteristics of the four methods and recent studies regarding these approaches.

Port-based approach: In the early days of the Internet, many application are distributed fixed network port numbers by the Internet Assigned Number Authority (IANA). The port-based approach extracts the port number in TCP or UDP header of packets of application traffic, and then compares extracted port number with IANA's registered list to classify network traffic. This method is very simple and effective in the early days of the Internet. However, with the common use of dynamic port and many services use same port number, for example, the voice over IP and video streaming use port 80 or 443 [14], port-based classification method is inaccurate in nowadays [12]. Currently, this method is applied to the encrypted traffic classification through combining with other methods.

Deep packet inspection (DPI): This method is also called signature-based method. This approach is to seek some signatures or information available in the payload of packets [15], and to compare the signatures with predefined patterns of specific application or traffic type to classify the network traffic. DPI method had obtained high classification performace in unencrypted network traffic. However, DPI approach is expensive because it need cost large labour power to maintain and update the predefined patterns [12]. In addition, with widespread use of encrypted technique and the enhancement of users' privacy protection consciousness nowadays, the DPI method is losing its ground.

Statistical and machine learning approach: This method is considered to be the most effective and efficient method. The method assumes that different applications or traffic types have unique statistical characterization to distinguish each other. The method usually leverages machine learning methods to classify and identify network traffic.

In [8], authors develop an online hierarchical classification system based on machine learning techniques. The classification system utilizes decision tree structure to decompose the multi-class problem into several binary problems, then Support Vector Machine and Neural Network are used to the binary classification. The classification system takes packet-level statistical values as traffic features, such as average packets size and average interarrival time etc., in the MAC layer. The classification system can categorize network traffic to the users' online activities (e.g., Browsing, chatting and BitTorrent). In [16], authors use the flow-based time-related features as traffic features, then use two well-known machine learning techniques (C4.5 and KNN) to characterize regular encrypted traffic and encrypted traffic tunneled through a Virtual Private Network into different categories, such as browsing and streaming, etc. In [17], authors proposed a session level flow classification (SLFC) approach to categorize network flows to a session. The approach first use the packet size distribution (PSD) to classify flows into the corresponding applications and then group the flows into sessions according to port locality. The approach calculate the PSD distance between the flow and pre-selected application, the flow with minimum PSD distance is classified as the application. In addition, port locality is used to group flows as sessions because an application often uses consecutive ports numbers within a session. In the [1], authors propose an application-level traffic classification method. The method uses unique payload size sequence (PPS) signatures generated by each application to identify application traffic. The PPS signature of each application consists of packet order, direction and payload size of the first N packets in a flow.

The feature extraction and selection approach play an essential role in network traffic classification, and it is important part of statistical and machine learning approach. The existing feature extraction and selection method includes manual, exhaustive and machine learning method. Manual feature extraction and selection method is time consuming and overly dependent on experts' experience. Exhaustive features search approach is computationally infeasible because of the huge number of feature set combinations. Feature extraction and selection using Machine learning method can optimize the classifier, but it can not be applied to online network traffic classification [18]–[20].

End-to-end method: With the application traffic encryption has become the standard practice nowadays, rendering encrypted traffic classification becomes a great challenge. And with the development of deep learning, it is applied to many fields and achieves significant results. Currently, there are only a few literatures apply deep learning method to encrypted traffic classification. In this section, we introduce several representative encrypted traffic classification methods. These methods are called as end-to-end method by a few scientific workers. One common characteristic of the end-to-end method is to utilize deep learning algorithms to integrates features extraction, features selection and classifier into a unified framework. End-to-end method can learn features and classify encrypted traffic automatically, it does not rely on handcrafted features. In [7] and [2], authors only use first 784 bytes of each flow or session, and select the traffic representation include Session+L7, Session+All, Flow+L7 and Flow+All. Then they convert these result files to gray images. Each byte of result file represents a pixel. They also can covert result files to IDX files directly. Finally, they uses one-dimensional convolution neural networks to classify application traffic and malware traffic into categories. In [16], authors propose an online IP traffic classification framework named Sequence-to-Image (Seq2Img), which base on online Convolution Neural Networks (CNNs). The framework employs a method base on compact non-parametric kernel embedding to convert early sequences of packets of network flows to images and then applies a CNN to classify generated images to different network applications. In [13], authors develop a framework called "Deep Packet" that comprises of two deep learning methods, CNN and stacked autoencoder NN. The framework applies stacked autoencoder NN to categorize the encrypted network traffic into major classes (e.g., FTP and P2P) and applies CNN to identify encrypted application traffic (e.g., Facebook, Skype). The method uses single packet as classification unit and identification unit.

III. Convolutional Neural Network

In recent years, with the development of deep learning, many deep learning methods are applied to many fields and achieves better results, such as CNN has been mainly applied in the domain of computer vision and natural language process [2].

People recognize the images through abstracting images layer by layer. People firstly understand the color and brightness of images, then understand the local detail features of images such as edge, corner and straight line, following understand more complex information and structure such as texture and geometry, and finally form the global concept of images. CNN is in line with our common sense in understanding images. CNN has outstanding ability in learning image features, it has unique method of extracting critical features of images, it has been widely applied in the field of computer vision, and it has achieve remarkable results in image recognition and image classification [21], [22]. CNN typically consists of few consecutive convolutional layers and pooling layers. CNN can automatically learns the features of different levels of images through convolution and pooling operations. The shallow convolutional layers can extract low-level features, such as edges and curves. The deeper convolutional layers can extract high-level abstract features. The convolution layers and pooling layers of CNN constantly extract and compress the features of images, and finally obtain a high level of features. In short, the original features are condensed step by step, and the final obtained features are more reliable. We can use the final obtained features to do a variety of tasks, such as classification, regression and so on [23].

Compared with prevailing artificial neural networks, a CNN has the following properties: First, the convolutional layers of a CNN are parameter sharing and sparse connectivity instead of full connectivity, which can decrease the number of parameter, reduce storage requirements for models and improve the statistical efficiency of the model. Second, pooling layers of a CNN can merely select salient features from small local receptive fields and further tremendously reduce the number of model parameters, it can improve spatial invariance of image to some extent. Third, fully-connected layers of CNN are used in the final stage, they integrates learned feature representations of previous layers together. The locally-connected convolutional layers and the pooling layers make CNN to deal with large-scale problems [12], [21], [24]–[26].

IV. End-to-end Framework

In this work, we propose a framework that consists of raw traffic preprocess phase, image conversion phase, CNN mode train and classification phase. The framework belongs to end-to-end method. Fig. 1 describes the all the phases and the specific processes of each phase. Following we will describe phases in detail in modules.

A. Dataset

In this work, we use public dataset ISCX. This dataset has rich classes and quantity. This dataset contains seven kinds of conventional encrypted traffic (e.g., P2P, VoIP) and seven kinds of encrypted traffic using Virtual Private Network (VPN) (e.g., VPN-P2P, VPN-VoIP). The function of the Virtual Private Network (VPN) is to establish a private network on a public network for encrypted communication. VPN gateway gets remote access through encrypting packets and converting destination address of packets [15]. In this dataset, each class consists of some different applications traffic in pcap format files captured by Wireshark and Tcpdump [16]. For example, the VoIP type consist of the traffic of Facebook, Hangouts, Skype and Voipbuster. The different pcap file is named according to the name of different application produced the packets (e.g., Skype, and Hangouts, etc.) and the specific activity the application was engaged during the capture session (e.g., voice call, chat, file transfer, or video call). For more details on the public dataset and the traffic generation process, please refer to [16]. In addition, we needed to label these pcap files as traffic types according to the description of the [16]. The Browsing is not the main activity, it is HTTPS traffic that was generated when users browsing or executing some activities using a browser. So we decided not to label the Browsing and VPN-Browsing. The detail of the dataset are listed in the Table I.

TABLE I
ISCX Dataset

Categories	Application
Chat	ICQ,AIM,Skype,Facebook,Hangouts
Email	SMPT,POP3,IMAP
File transfer	Skype,FTPS,SFTP
VoIP	Facebook,Skype,Hangouts,Voipbuster
P2P	Torrent
Streaming	Vimeo,Youtube,Netflix,Spotify
VPN-Chat	ICQ,AIM,Skype,Facebook,Hangouts
VPN-Email	SMPT,POP3,IMAP
VPN-File transfer	Skype,FTPS,SFTP
VPN-VoIP	Facebook,Skype,Hangouts,Voipbuster
VPN-P2P	Bittorrent
VPN-Streaming	Vimeo,Youtube,Netflix,Spotify

B. Data Preprocessing

Data preprocessing includes two steps: traffic split, traffic clear.

Step 1 (Traffic Split): This step splited continuous raw traffic to discrete sessions. We used network packet analysis tool Wireshark to filter out the sessions based on selected features. The output data are saved as CSV format files.

Step 2 (Traffic Clear): Firstly, we employed the session as our traffic unit. Flow is defined as the set of packets with the same 5-tuple, i.e. source IP, source port, destination IP, destination port, transport-level protocol and are listed in time order. Session is defined as the flow including the forward and backward packets [27]–[29]. The lifetime of sessions is define as the following policies: TCP sessions are terminated after first capture the packet with FIN/RST. If there is not the packet with FIN/RST in the TCP sessions, we terminate the sessions after timeout. UDP sessions are terminated only after timeout.

Secondly, we must remove some small sessions. As we need to utilize first few payload sizes of sessions to classify the encrypted network traffic, we remove the sessions that less than three non-zero payload sizes in the forward or the backward direction, whether it is TCP sessions or UDP sessions. In addition, since the dataset is captured in a real-world emulation, it contains some assisted sessions which are useless for us and should be removed, such as Domain Name Service (DNS) packets that are used for hostname resolution. NBNS packets provide the methods of hostname and address mapping on name accessed networks based on NetBIOS names.

Finally, as to TCP sessions, since the SYN packets (Three-way handshakes), ACK packets and the FIN packets (Four waves) contain no payload in the transport layer, They are used to establish and terminate a connection, they carry no information regarding the application generated them, so we remove them. In addition, when the capture tool catches same packets twice, and it catches the ACK of the first packet, the capture tool judges that the initial packet was not actually lost, so it labels the second packet Spurious Retransmission. Unreachable packets and spurious retransmission packets of sessions are redundant packets captured by the capture tools. We remove the packets with Unreachable and Spurious Retransmission in the sessions.

C. Image-based Method

We get the sessions after processing the raw network traffic. Considering some application traffic include TCP and UDP sessions (e.g., file), as to TCP sessions, the packets with zero payload size are often confirmation packets or control packets, which carry no information regarding the application generated them, so we ignore them. Thus we only consider the non-zero payload size of sessions. We covert first few non-zero payload sizes of session to binary data, and joint these binary data. Then we represent a byte as integer from 0 to 255. As a consequence, the converted gray image can reflect the intrinsic characteristics of session. As to non-VPN traffic, we convert

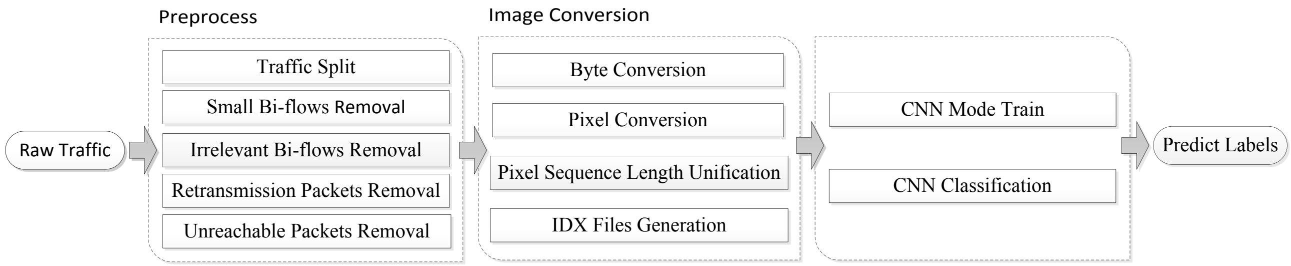

Fig. 1. End-to-end framework

the first 90 non-zero payload sizes of each session to a gray image. And as to VPN traffic, we covert the first 20 non-zero payload sizes of each session to a gray image. The reason of choosing two values will be given in section V.

As CNN achieves very high classification performance in image classification. And the data fed into CNN must be uniform, so we decide the size of gray images converted by sessions is 28*28 (784 bytes). If the length of pixel sequence of converted image less than 784 bytes, we pad zero in the end to complement it to 784 bytes. If the length of pixel sequence of converted image more than 784 bytes, it is trimmed to 784 bytes. Finally, we can covert the file that contains all the pixel sequences into IDX format files. The IDX file format is a simple format for vectors and multidimensional matrices of various numerical types [30].

D. CNN Model

We use Convolutional Neural Network (CNN) to classify the converted gray images. CNN is a kind of feedforward neural network with convolution calculation and deep structure, and it is one of the representative algorithms of deep learning. CNN is good for processing data with like grid structure, such as time-series data and image data. Convolution neural network has excellent performance in many application fields, such as image recognition [21], [31], [32], video analysis [26], [33] and natural language processing [34]. Specially, 1D-CNN is commonly used to data like sequential data and natural language processing, 2D-CNN is commonly used to data like image and audio spectrograms. 3D-CNN is commonly used to data like video and volumetric images.

A typical CNN architecture consists of input layer, output layer, hidden layer. Hidden layer contains convolutional layer, pooling layer, incentive layer, fully connected layer. The input layer preprocesses the initial data. Convolutional layer extracts features of data. Pooling layer selects extracted features of convolutional layer, reduces the number of parameters and prevents overfitting. Incentive layer maps the output of the convolution layer nonlinearly. The full connected layer integrates feature representations and output layer outputs the predicted results [12].

We use TensorFlow platform to built 1D-CNN. TensorFlow is an end-to-end open source platform for machine learning (ML). It has a comprehensive, flexible ecosystem of tools, libraries and community resources that lets researchers push the state-of-the-art in ML and developers easily build and deploy ML powered applications. We compare the classification performance of 1D-CNN with that of 2D-CNN through performing the experiments, the classification performance of 1D-CNN is slightly better than that of 2D-CNN. We describes the main parameters of each layer of the 1D-CNN model in Table II.

TABLE II
THE MAIN PARAMETERS OF 1D-CNN MODEL

Layer	Operation	Input	Filter	Stride	pad	Output
1	Conv+Relu	784*1	25*1	1	same	784*32
2	maxpool	784*32	3*1	3	same	262*32
3	Conv+Relu	262*32	25*1	1	same	262*64
4	maxpool	262*64	3*1	3	same	88*64
5	Full connect	88*64	–	–	none	1024
6	softmax	1024/6/6	–	–	none	6

V. EXPERIMENT

To evaluate the classification performance of our proposed method, We use three metrics: precision (Pr), recall (Rc) and f1 score (F1). These evaluative metrics are described mathematically as follows:

$$Pr = \frac{TP}{TP+FP}, Rc = \frac{TP}{TP+FN}, F1 = \frac{2*Pr*Rc}{Pr+Rc} \quad (1)$$

A. Performance Comparison

In the Table III, we compare our proposed method with the methods [13], [35]. The two methods use deep learn algorithms, hence they belong to end-to-end method. In the Table III, the precision, recall and F1 score are average value of that of conventional traffic and VPN traffic. From the Table III, we can know that the classification performance of our proposed method is better than two methods [13], [35] in both regular traffic and VPN traffic. Table III reports that the precision, recall and F1 score are 5.65%, 4.64% and 5.64% higher respectively than the method [13] and 7.65%, 7.64% and 7.64% higher respectively than the method [35].

TABLE III
PERFORMANCE COMPARISON (%)

Work	Method	Precision	Recall	F1 Score
This Paper	1D CNN	98.65	98.64	98.64
Ref. [35]	CNN+LSTM	91	91	91
Ref. [13]	1D CNN	93	94	93

B. Selection of The Length of Subflow

In the Fig. 2 and Fig. 4, the X-axis is the length of the subflows, the Y-axis is the percent value of precision, recall and F1 score. As we can tell from the Fig. 2 and Fig. 4, different lengths of subflow affect the classification performance whether conventional traffic or VPN traffic. Further, for the conventional traffic, the maximum difference of f1 score with different lengths of subflow can be up to 14.42%. For the VPN traffic, the maximum difference of f1 score with different lengths of subflow can be up to 5.8%. In addition, as shown in Fig. 2, in the conventional traffic, our proposed method can obtain much accurate by using first 90 payload sizes of session. As shown in Fig. 4, in the VPN traffic, our proposed method can obtain better classification result by using first 20 payload sizes of session. This why our proposed method converts the first 90 payload sizes of each session to gray image in conventional traffic and convert the first 20 payload sizes of each session to gray image in VPN traffic.

Futher, the Fig. 3 presents the classification performance of our proposed method in all classes when converting the first 90 payload sizes of each session to gray image in conventional traffic. The Fig. 5 presents the classification performance of our proposed method in all classes when converting the first 20 payload sizes of each session to gray image in VPN traffic. As Fig. 3, Fig. 5 and Table IV show, our proposed method achieves high accuracy in P2P traffic and do not work well in chat traffic whether conventional traffic and VPN traffic.

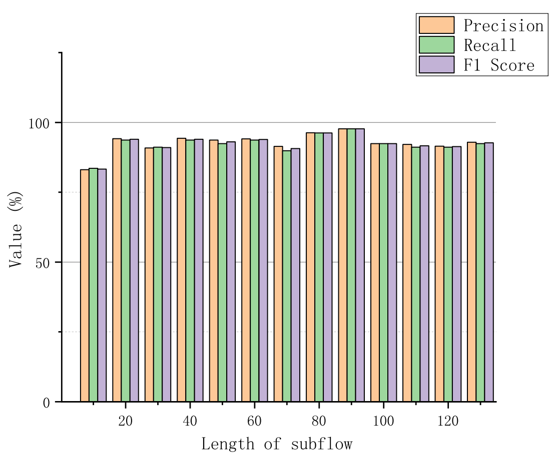

Fig. 2. the classification performance of conventional traffic with different lengths of subflow

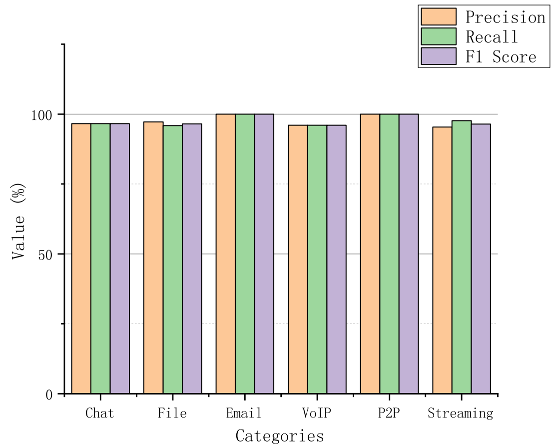

Fig. 3. the classification performance of categories of conventional traffic

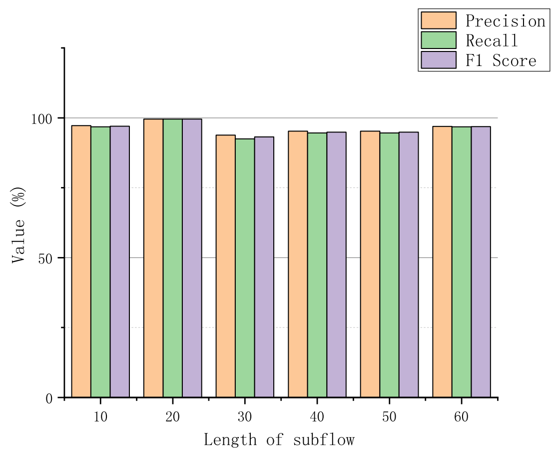

Fig. 4. the classification performance of VPN traffic with different lengths of subflow

TABLE IV
CLASSIFICATION PERFORMANCE (%)

Class	Pre	Rec	F1	Class	Pre	Rec	F1
Chat	96.56	96.56	96.56	VPNChat	100	95.24	97.56
File	97.2	95.86	96.53	VPNFile	98.41	100	99.2
Email	100	100	100	VPNEmail	100	97.73	98.85
VoIP	96	96	96	VPNVoIP	100	100	100
P2P	100	100	100	VPNP2P	100	100	100
Stream	95.35	97.62	96.47	VPNStream	100	100	100

VI. DISCUSSION

Compared with the traditional methods, our proposed image-based encrypted traffic classification method can automatically extract features, select features of encrypted network traffic and classify encrypted network traffic. The method does

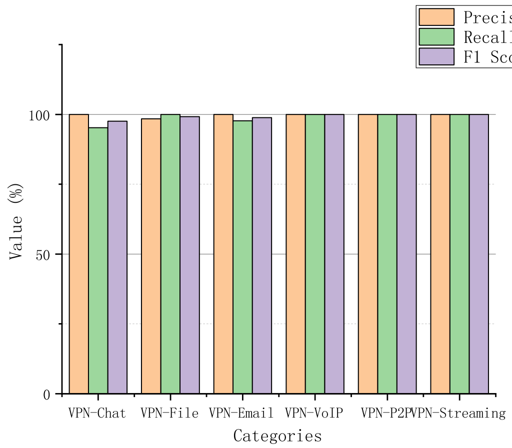

Fig. 5. the classification performance of categories of VPN traffic

not rely on experts' experience. In addition, our proposed method only utilizes first few non-zero payload sizes of sessions, thus they can be easily implemented to classify the encrypted network traffic. However, there are a few problems on the proposed method need to further study in the future work. Firstly, we geted the length of subflows in a coarse-grained way, we plan to study an algorithm to get the length of subflows that improves classification performance in the future work. Secondly, we achieved the length of subflows that covert into gray images, but the length of subflows of regular traffic is different from that of subflow of VPN network traffic. As the gap between the two is quite large, we will study how to deal with the problem in future work. Finally, our proposed method only classify the six common traffic types, we want to focused on studying to classify more network traffic types in the future work.

VII. CONCLUSION

In this paper, we proposed an image-based classification method, this method is end-to-end method. The method applies deep learning algorithm to network traffic classification. The proposed method can automatically extract and select features of encrypted traffic and classify encrypted network traffic. Compared to traditional divide-and-conquer methods, our proposed method does not depend on handcraft features. The proposed method achieves outstanding performance in encrypted traffic classification. The experiment results show that the proposed method achieves a significant improvement of classification performance than the state-of-the-art on the public ISCX dataset. In addition, we studied the effect of length of subflow on classification performance. We achieved the length of subflow to improve the classification performance instead of using first 784 bytes or 900 bytes of session arbitrarily. As the proposed method only use first few non-zero payload sizes of sessions, the method is very light-weight. The dataset ISCX consist of seven kinds of conventional traffic and seven kinds of VPN traffic, these classes are common traffic. The applications of each class are popular. Therefore, the proposed method can be applied to in real world.

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